

EF5342 Financial Systems, Markets and Instruments

Week 1 — Overview of the Financial System

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City University of Hong Kong

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About Me

- Ph.D. in Finance, HKUST; CFA charterholder since 2021
- Associate Professor of Finance; prior to joining CityU, taught at Singapore Management University
- Research in behavioral finance, asset pricing, and ESG, published in top journals
- Consultant to quant funds (Magnum Research Global, Heritage Capital Management)
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Logistics

- Check Canvas regularly for announcements and downloads
- One TA per section — contact your section's TA for all course matters

Final exam: one two-sided A4 cheat sheet (name on top) allowed; calculator permitted per university regulations.

Assessment

- 20% individual assignments (two sets, work independently)
- 30% group project
- 50% final exam (MCQs + calculations, similar to assignments)

Group Project (30%)

- Analyze financial market instruments or institutions; requires collecting and analyzing financial data
- Two topics announced in Week 2 — your group picks one
- Form groups by Week 3 (max 7 students; TA re-shuffles if needed; email TA your group composition)
- Report (20%) due Week 10, Sunday — email to me
- Presentation (10%): 15 minutes + 5 minutes Q&A; all members present; Q&A not dominated by one member

My Teaching Style

- **Broad and practical:** real-world examples; figures illustrating trends and patterns; a recent case or news item opening every class
- **Goal:** understand *why* financial markets behave as they do, and how to interpret financial news
- **Research-oriented:** as a research-track professor, I bring relevant academic research into class; the project teaches you to put every claim to the test with data

Preview: The Robot-Startup Story

Setup: You want to start a business manufacturing a household cleaning robot, but you have no funds. At the same time, Walter has money he wishes to invest for retirement.

Question: If the two of you could get together, both needs could be met. *How does that happen?*

As simple as this example is, it highlights the importance of financial markets and financial intermediaries in our economy — we need to understand their structure and operations before we can appreciate their role.

Roadmap for Today

We examine the role of the financial system in an advanced economy: the effects of financial markets and institutions on the economy, their structure and operations.

- Function of financial markets
- Structure of financial markets
- Internationalization of financial markets
- Function of financial intermediaries: indirect finance
- Types of financial intermediaries
- Regulation of the financial system

Function of Financial Markets

Channels funds from those **without** investment opportunities (“lender-savers”) to those **with** them (“borrower-spenders”) — improving economic efficiency.

Why it matters: save \$1,000 with no financial markets and you earn no return — might as well put the money under your pillow. But a carpenter who can buy a new saw with that money (raising her productivity) will pay you interest for it.

Financial markets produce an **efficient allocation of capital** and let consumers **time their purchases** better.

Who Are the Lender-Savers and Borrower-Spenders?

Lender-Savers (surplus units)

- Households
- Firms
- Government
- Foreigners

Borrower-Spenders (deficit units)

- Firms
- Government
- Households
- Foreigners

The same four sectors appear on both sides — who is which depends on circumstances, not identity.

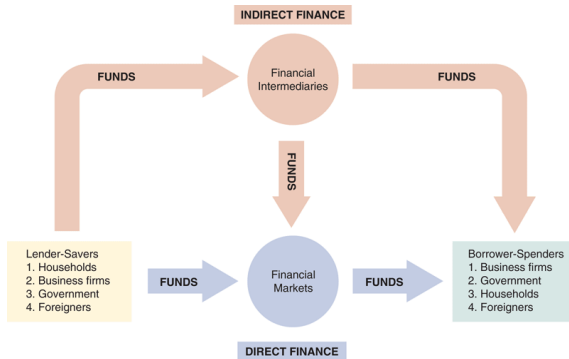
Two Segments: Direct vs. Indirect Finance

Direct finance Borrowers borrow *directly* from lenders in financial markets by selling securities — claims on the borrower's future income or assets

Indirect finance Borrowers borrow *indirectly* via financial intermediaries, which source both loanable funds and loan opportunities

Next: the map of how funds actually flow through the system.

Flows of Funds Through the Financial System



The arrows show that funds flow from lender-savers to borrower-spenders via two routes: *direct finance*, in which borrowers borrow funds directly from financial markets by selling securities, and *indirect finance*, in which a financial intermediary borrows funds from lender-savers and then uses these funds to make loans to borrower-spenders.

Figure: Mishkin, Figure 2.1

Structure: Debt vs. Equity Markets

Markets can be classified along several (not mutually exclusive) dimensions.

Debt markets

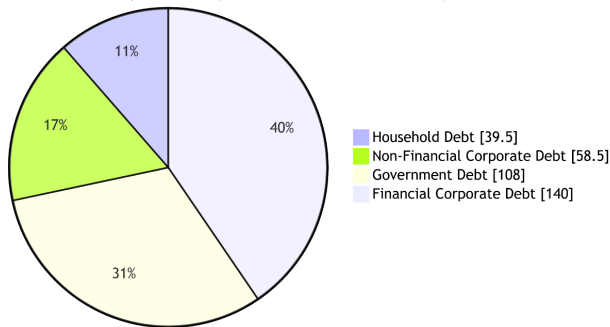
- Money market: maturity < 1 year
- Intermediate term: in between
- Long term: maturity > 10 years
- Global total $\approx$ \$353 trillion (Q1 2026, IIF Global Debt Monitor)

Equity markets

- Ownership claim in the firm
- Pays dividends — in theory, forever
- Voting rights
- Global market cap $\approx$ \$158 trillion (end-2025, WFE/SIFMA)

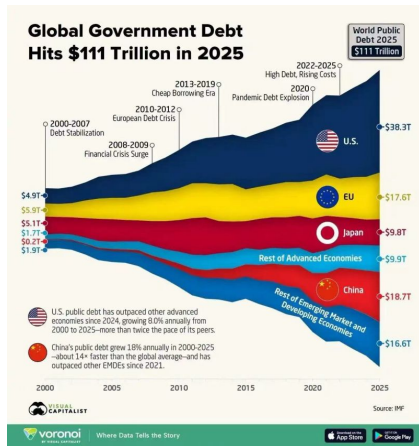
The Global Debt Market

Debt Breakdown by Sector (Q3 2025, ~\$346 Trillion)



Source: Institute of International Finance (IIF), Global Debt Monitor (chart data as of Q3 2025)

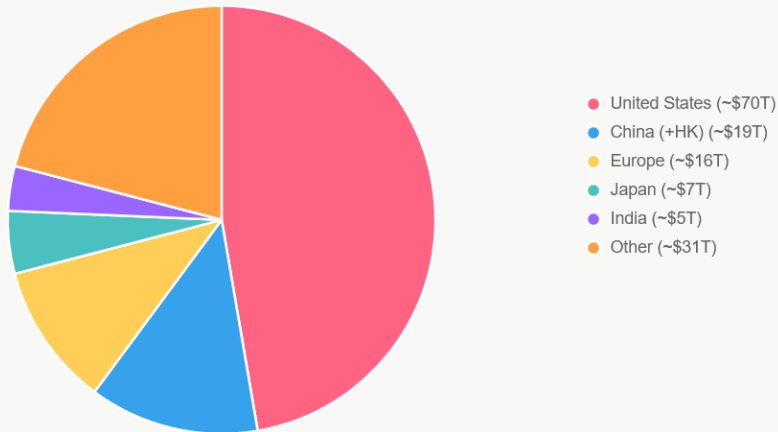
Global Debt: Sector Composition



Source: Institute of International Finance (IIF), Global Debt Monitor (chart data as of 2025)

The Global Equity Market

Global Equity Markets Breakdown by Key Regions/Countries (End-2025, ~\$148T Total)



Source: World Federation of Exchanges (WFE); chart data as of end-2025

Primary vs. Secondary Markets

Primary market New security issues sold to initial buyers; typically involves an investment bank that *underwrites* the offering

Secondary market Previously issued securities bought and sold (NYSE, Nasdaq, HKEX); involves brokers and dealers

Firms get no new capital from the secondary market, yet it serves two functions:

- **Liquidity** — easy to buy and sell the company's securities
- **Price discovery** — establishes prices useful for valuing the firm

Secondary Markets: Exchanges vs. Over-the-Counter

Exchanges Trades conducted in central locations — e.g., NYSE, HKEX

Over-the-counter (OTC) Dealers at different locations buy and sell; best example is the market for U.S. Treasury securities

Classifying Markets by Maturity

Money market Short-term instruments, maturity < 1 year

Capital market Long-term instruments, maturity > 1 year, plus equities (no finite maturity)

Internationalization: Foreign Bonds and Eurobonds

Foreign bond — issued in a domestic market by a foreign entity, denominated in the domestic currency. *Example:* a Japanese company issues a USD-denominated bond in the U.S.

Eurobond — issued in a currency that is not native to the country or market where it is issued. *Example:* a Japanese company issues a USD-denominated bond in London.

- Over 80% of new bonds are Eurobonds
- Attractive for **regulatory arbitrage**

Case: “Wonton Bonds” in Hong Kong

Foreign bonds denominated in HKD, issued by non-HK entities — mainly supranational / multilateral development banks (AIIB, IFC/World Bank, ADB) and some foreign banks and corporates.

- Often sustainable / social / green labeled
- Publicly syndicated — broader investor access than private deals
- Attractive in low-rate environments for diversification away from USD
- Help HK banks and treasuries deploy excess HKD liquidity into high-quality assets

The Eurocurrency Market

An international, largely **unregulated** network where currencies are deposited, lent, and borrowed *outside their country of origin*.

- **Eurodollars**: U.S. dollars deposited in (say) London; **Euroyen**: yen deposited in New York
- Chosen over domestic banks for lower borrowing rates and higher deposit rates, situationally
- Why? Fewer regulatory requirements, different tax laws, and typically no interest-rate caps

Eurocurrency Market: Pros and Cons

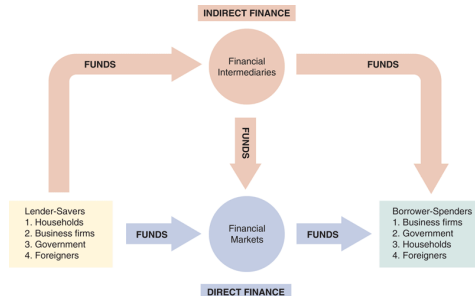
Advantage	Disadvantage / Risk
Lower borrowing costs: fewer regulations allow competitive rates	Regulatory gaps: transnational nature complicates oversight
Higher deposit returns: banks can offer better rates	No lender of last resort: deposits lack a central-bank safety net
Efficient fund management: hedging currency risk, global cash management	Increased systemic risk: size and interconnectedness amplify shocks
Key source of trade finance: facilitates over 90% of international trade	

World Stock Market Indexes

Published daily (e.g., finance.yahoo.com). The most important:

- **DJIA** — 30 largest U.S. public corporations
- **S&P 500** — 500 largest U.S. companies
- **Nasdaq Composite** — all Nasdaq stocks; most U.S. technology names
- **FTSE 100** — 100 largest UK companies (LSE)
- **DAX** — 30 largest German companies (Frankfurt)
- **CAC 40** — 40 largest French companies (Euronext Paris)
- **Hang Seng** — largest HK-listed companies
- **Straits Times** — 30 largest companies (SGX)

Indirect Finance: Financial Intermediation



The arrows show that funds flow from lender-savers to borrower-spenders via two routes: *direct finance*, in which borrowers borrow funds directly from financial markets by selling securities, and *indirect finance*, in which a financial intermediary borrows funds from lender-savers and then uses these funds to make loans to borrower-spenders.

Instead of savers lending directly to borrowers, an intermediary (such as a bank) acts as the middleman: it **obtains funds from savers**, then **makes loans and investments to borrowers**.

Intermediation Dominates Direct Finance

Financial intermediation is the **primary** means of moving funds from lenders to borrowers.

- More important than direct finance in capital markets
- In Germany and Japan, intermediary financing exceeds capital-market financing **10-fold**
- Exists because direct finance suffers from: **transaction costs**, **risk sharing**, and **asymmetric information**

Reason 1: Transaction Costs and Liquidity Services

FIs make profits by **reducing transaction costs**, via expertise and **economies of scale**. Low transaction costs also enable **liquidity services** — making transactions easier for customers:

- Checking accounts let depositors pay bills easily
- HK depositors earn interest on savings yet can convert to goods and services at any time

Reason 2: Risk Sharing and Asset Transformation

FIs reduce investors' exposure to **unsystematic risk**:

- FIs create and sell low-risk assets to one party, buy higher-risk assets from another — **asset transformation**: risky assets turned into safer assets
- E.g., banks lend to risky firms yet issue deposits that are almost risk-free
- Possible due to economies of scale: buy a range of assets, pool them, sell rights to the diversified pool

Reason 3: Asymmetric Information

One party lacks crucial information about the other. Two fronts:

Adverse selection *Before* the transaction: the potential borrowers most eager to borrow are the ones most likely to produce adverse outcomes (cf. unhealthy people most eager to buy insurance)

Moral hazard *After* the transaction: the borrower has incentives to engage in undesirable activities that make repayment less likely (cf. insured people taking more risks); a conflict of interest

FIs have expertise to mitigate both — and profit from it. How they do it: throughout this course.

Economies of Scope — and Conflicts of Interest

Economies of scope: FIs lower the cost of information by reusing it across services — bank accounts, mortgages, insurance, retirement savings. *Example:* a mortgage applicant's banker can suggest a homeowner checking account, a credit card for moving costs, and home insurance.

But multiple services create **conflicts of interest** — one area of the FI hiding information from another:

- Selling for commissions and quotas
- Promoting proprietary “in-house” products
- Data-driven exploitation

This can make financial markets *less* efficient and reduce consumer welfare.

Types of Financial Intermediaries

Type of Intermediary	Primary Liabilities (Sources of Funds)	Primary Assets (Uses of Funds)
Depository institutions (banks)		
Commercial banks	Deposits	Business and consumer loans, mortgages, U.S. government securities, and municipal bonds
Savings and loan associations	Deposits	Mortgages
Mutual savings banks	Deposits	Mortgages
Credit unions	Deposits	Consumer loans
Contractual savings institutions		
Life insurance companies	Premiums from policies	Corporate bonds and mortgages
Fire and casualty insurance companies	Premiums from policies	Municipal bonds, corporate bonds and stock, and U.S. government securities
Pension funds, government retirement funds	Employer and employee contributions	Corporate bonds and stock
Investment intermediaries		
Finance companies	Commercial paper, stocks, bonds	Consumer and business loans
Mutual funds	Shares	Stocks, bonds
Money market mutual funds	Shares	Money market instruments

Source: Federal Reserve Flow of Funds Accounts; www.federalreserve.gov/releases/Z1/.

Figure: Mishkin, Table 2.1 — Primary Assets and Liabilities of Financial Intermediaries

Principal U.S. Financial Intermediaries by Asset Size

Type of Intermediary	Value of Assets (\$ billions, end of year)			
	2000	2010	2020	2021
Depository institutions (banks)				
Commercial banks, savings and loans, and mutual savings banks	7,687	15,580	21,644	23,587
Credit unions	441	912	1,823	2,041
Contractual savings institutions				
Life insurance companies	3,136	5,176	9,420	9,863
Fire and casualty insurance companies	862	1,242	2,879	3,100
Pension funds (private)	4,355	4,527	12,035	13,169
State and local government retirement funds	2,293	2,661	9,513	9,818
Investment intermediaries				
Finance companies	1,140	1,439	1,607	1,671
Mutual funds	4,435	7,935	19,563	22,209
Money market mutual funds	1,812	2,755	4,766	5,206

Depository Institutions (Banks)

Accept deposits and make loans — commercial banks and thrifts.

Commercial banks Raise funds via current accounts, savings accounts, and time deposits; lend to commercial, consumer, and mortgage borrowers. The largest FI group, with the most diversified asset portfolios

Thrifts: S&Ls, mutual savings banks, credit unions Raise funds via savings, time, and checkable deposits; mostly mortgage and consumer lending. Mutual savings banks and credit unions are depositor-owned; credit union members typically share a common bond (e.g., a company's workers)

Contractual Savings Institutions

Acquire funds at **periodic intervals on a contractual basis**; payout requirements are fairly predictable.

Life insurance Funds from premiums; can hold less liquid, longer-maturity corporate bonds and mortgages — actuarial payouts are predictable

Fire and casualty insurance Funds from premiums; must hold mostly liquid government and corporate securities — losses are hard to predict

Pension and government retirement funds Contributions from employee and employer payrolls; invest in corporate securities; pay retirement income via annuities

Other Intermediaries

Finance companies Sell commercial paper, issue bonds and stocks; lend to consumers (durables) and small businesses

Mutual funds Sell shares to individual investors; purchase diversified portfolios of stocks and bonds

Money market mutual funds Sell checkable deposit-like shares; hold liquid, safe short-term money-market instruments

Hedge funds Large minimum investments (\$100,000+), long lock-ups, few regulations; leverage and derivatives across asset classes

Investment banks Advise on securities issuance, underwrite offerings, M&A advisory, deal in security markets

U.S. Regulatory Agencies

Regulatory Agency	Subject of Regulation	Nature of Regulations			
Securities and Exchange Commission (SEC)	Organized exchanges and financial markets	Requires disclosure of information; restricts insider trading	State banking and insurance commissions	State-chartered depository institutions	Charter and examine the books of state-chartered banks and insurance companies; impose restrictions on assets they can hold; and impose restrictions on branching
Commodities Futures Trading Commission (CFTC)	Futures market exchanges	Regulates procedures for trading in futures markets	Federal Deposit Insurance Corporation (FDIC)	Commercial banks, mutual savings banks, savings and loan associations	Provides insurance of up to \$250,000 for each depositor at a bank; examines the books of insured banks; and imposes restrictions on assets they can hold
Office of the Comptroller of the Currency	Federally chartered commercial banks and thrift institutions	Charters and examines the books of federally chartered commercial banks and thrift institutions; imposes restrictions on assets they can hold	Federal Reserve System	All depository institutions	Examines the books of commercial banks that are members of the system; sets reserve requirements for all banks
National Credit Union Administration (NCUA)	Federally chartered credit unions	Charters and examines the books of federally chartered credit unions and imposes restrictions on assets they can hold			

Mishkin, Table 2.3 — Principal Regulatory Agencies of the U.S. Financial System

Regulatory Agencies in Hong Kong

- HKMA — Hong Kong Monetary Authority De facto central bank: banking stability, monetary policy (Linked Exchange Rate System), prudential supervision
- SFC — Securities and Futures Commission Market integrity, investor protection, supervision of securities and futures activities
- IA — Insurance Authority Prudential supervision of insurers; conduct regulation of intermediaries
- MPFA — MPF Schemes Authority Regulates and supervises the Mandatory Provident Fund system

Why Regulate?

- 1. Increase investor information.** Asymmetric information causes adverse selection and moral hazard, hindering efficient market operation and driving investors away. The SEC requires disclosure of sales, assets, and earnings, and restricts trading by corporate insiders — increasing information available to investors.
- 2. Ensure soundness of FIs.** Depositors cannot easily assess whether an institution is sound; doubts can trigger withdrawals from *both sound and unsound* institutions — a financial panic (bank run) with large public losses and serious economic damage.

Six Types of Regulation

To protect the public from financial panics, governments impose:

- ① Restrictions on entry
- ② Disclosure
- ③ Restrictions on assets and activities
- ④ Deposit insurance
- ⑤ Limits on competition
- ⑥ Restrictions on interest rates

Each in turn on the next frames.

Regulation 1–3: Entry, Disclosure, Assets

Restrictions on entry — tight rules on who may set up an FI. In the U.S., a charter from state or federal government requires impeccable credentials and substantial initial funds. In HK, a license from the HKMA or SFC under stringent criteria.

Disclosure — strict reporting: bookkeeping must follow strict principles, books are periodically inspected, certain information must be made public.

Restrictions on assets and activities — restrict FIs from engaging in certain risky activities and from holding risky assets (or more of them than is prudent), so depositors' funds are safe.

Regulation 4: Deposit Insurance

Government insures deposits against FI failure.

	United States (FDIC)	Hong Kong (DPS)
Coverage	USD 250,000 per account (since 2008)	HK\$800,000 per depositor per bank (since 1997)
Applies to	Commercial and mutual savings banks	Licensed banks

Regulation 5–6: Competition and Interest Rates

Limits on competition. Evidence that competition among FIs promotes harmful failures is weak — yet many restrictions exist. U.S. banks were once barred from branching across states; in HK a bank must inform the HKMA and obtain a no-objection before opening a branch.

Restrictions on interest rates. Instituted on the belief that unrestricted deposit-rate competition encouraged Great-Depression-era bank failures; later evidence did not support this view, and the restrictions have been abolished.

A Third Reason: Improve Monetary Control

Banks play a central role in determining the money supply, which affects many aspects of the economy.

- **Reserve requirements:** depository institutions must keep a fraction of deposits in accounts with the Fed
- This helps the Fed exercise more precise control over the money supply

Many countries with similar economic systems — Japan, Canada, Western Europe, China, Singapore — implement regulation consistent with the U.S. model:

- Financial reporting for FIs is required
- Financial intermediaries are heavily regulated

However, U.S. banks are said to be *more* regulated than banks elsewhere along the dimensions of branching and services.

Summary

- Function of financial markets Flow of funds through the system; the role of intermediaries
- Structure of financial markets By instrument type, purpose, organization, and time horizon
- Internationalization Debt and equity markets in the international setting
- Function of financial intermediaries Reducing transaction costs, facilitating risk sharing, reducing information asymmetry
- Types of intermediaries The landscape, examined further in later weeks
- Regulation of the financial system The agencies overseeing institutions and markets